

Schoenfield on why higher order defeat is special

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3 April 2023

1. Bayesian Defeatism

HYPOXIA: You're a cargo pilot who was hired to fly from London to Philadelphia to deliver some goods. While in the air you get a message from your employer. They inform you that, due to some recent developments, it would be optimal for the goods to be delivered to [City] rather than Philadelphia. Unfortunately, they can't afford a refueling stop, but they're wondering whether you have enough fuel to safely make it to [City]. At this point [t_0] your credence that you have enough fuel to make it to [City] (call this proposition L) is [x].

You know that you can easily gather some evidence, which (together with your background knowledge, K) will support either a high credence in L or an equally high credence in $\sim L$ (depending of course on what type of evidence you receive). So, at 12pm you gather some evidence by looking at your gauges dials and maps (call the evidence you get E) which in fact supports L. You engage in some complex reasoning and [at t_1] you rationally become confident that L on its basis.

You get a call from ground control at 12:05pm [= t_2]. They alert you that for the last five minutes you've been flying at an altitude which puts you at risk for hypoxia: a condition that impairs one's ability to engage in the kind of reasoning you just did, even though everything seems normal from the inside. Pilots flying at your altitude, you are told, reason in a way that is no better chance [sic]. Call the information you get from ground control "H."

Defeatist Verdict: Your credence in L after you learn H, *should* go back to [x] (or thereabouts).

- Some earlier work suggested (to some) a view where your credence should go to 0.5, irrespective of your priors. But this seems implausible.

Miriam's questions: is there a way of getting the Defeatist Verdict that's compatible with standard Bayesian modeling assumptions? If not, why not?

- In some versions of this story, it's stipulated that actually you do not have hypoxia (or are not at the high altitude). I find this addition helpful, since it's a bit hard to say how someone with this extreme sort of hypoxia could do anything 'rationally' at all. The Defeatist Verdict doesn't change.
- What does it mean to 'reason in a way that is no better than chance'? Maybe something like this: there is zero *correlation* (according to the objective chances at t_0) be-

tween the content of the new evidence you get by looking at the gauges etc. and the credences about L you end up with at t_1 . One way for this to be true: pilots at this altitude who look at their gauges and maps *always* end up confident they have enough fuel to make it to a city whenever their prior credence they they could make it there was above 0.3. Another way: an internal coin toss determines whether they are confident that they can or confident that they can't.

2. Why Defeatism is hard to reconcile with Bayesianism

It's natural to suppose that your priors P_0 (at t_0) were such that H is *independent* of L and E—in particular, $P_0(H | L \wedge E) = P_0(H | \neg L \wedge E)$.

- After all, H is just a proposition about your own physiological state at a particular time (t_1), intuitively no more relevant to L and E than propositions about anyone else's hypoxia, or your own hypoxia at some far-future time.
- Some Bayesian Defeatists might challenge this independence claim. But this doesn't strike me as a plausible way forward.

It follows from this that $P_0(L | H \wedge E) = P_0(L | E)$.

But then, if you update by conditionalizing first on E (from t_0 to t_1) and then on H (from t_1 to t_2), your credence in L will not change from t_1 to t_2 , which conflicts with Defeatism.

However, this doesn't immediately doom the application of a Bayesian model to defeatism. The Bayesian defeatist can say either that the transition from from t_0 to t_1 involves conditionalizing on some further relevant evidence that goes beyond E, or that the transition from from t_1 to t_2 involves conditionalizing on some further relevant evidence that goes beyond H.

- Which option should the Bayesian Defeatist adopt? It seems to me most plausible to think that even at t_1 , your credence in L conditional on H should be around x . After all you might regard hypoxia as a distant possibility and explicitly consider the supposition that you have it.
- If so, the transition from t_0 to t_1 can be modeled straightforwardly as conditionalization on H. The Bayesian Defeatist will have to say that between t_0 and t_1 , you learn, and conditionalize on, some other (relevant) proposition J in addition to E.
- Plausibly, J will be some psychological proposition about your response to E. Miriam suggests that it might be a proposition about what you "judge", such as the proposition that you judge L.

3. Miriam's central argument against Bayesian Defeatism

The argument involves referring to two principles.

RatEL: believing L [i.e. having high credence in L] is rational in response to E [+background knowledge K].

- We can gloss this [tendentiously!] as: ‘the unique rational ur-prior function p is such that $p(L \mid E \wedge K)$ is high’.

BRIDGE: Let p be the rational ur-prior and let $p^* = p(\cdot \mid p \text{ is rational})$. Then $p = \text{Exp}_p(p^*)$ (that is, p equals the expectation of p^* relative to p) and $p(\cdot \mid E) = \text{Exp}_{p(\cdot \mid E)}(p^*(\cdot \mid E))$.

- I think this can be spelled out as follows, assuming discreteness for simplicity. If p is the RUP (rational ur-prior), then
 - a) for any A , $p(A) = \sum_q p(q \text{ is the RUP}) q(A \mid q \text{ is the RUP})$.
 - b) for any A and E , $p(A \mid E) = \sum_q p(q \text{ is the RUP} \mid E) q(A \mid q \text{ is the RUP} \wedge E)$.
- We can note that BRIDGE(a) is equivalent to BRIDGE(b), and also—given the assumption that $p(\text{there is exactly one RUP}) = 1$ —to

UR-PRIOR NEW RATIONAL REFLECTION: $p(A \mid q \text{ is the RUP}) = q(A \mid q \text{ is the RUP})$, where A is any proposition, p is the RUP, and $p(q \text{ is the RUP}) > 0$.

Proof: from BRIDGE(a) to UNRR: $p(A \wedge q \text{ is the RUP}) = \sum_{q'} p(q' \text{ is the RUP}) q'(A \wedge q \text{ is the RUP} \mid q' \text{ is the RUP}) = p(q \text{ is the RUP}) q(A \mid q \text{ is the RUP})$.

From UNRR to BRIDGE(b): $p(AE) = \sum_q p(AE \wedge q \text{ is the RUP}) = p(E) \sum_q p(q \text{ is the RUP} \mid E) p(A \mid q \text{ is the RUP} \wedge E) = p(E) \sum_q p(q \text{ is the RUP} \mid E) q(A \mid q \text{ is the RUP} \wedge E)$.

From BRIDGE(b) to BRIDGE(a): let $E = \top$.

Referring to these principles, we can state an argument against Bayesian Defeatism as follows, focusing for simplicity on the $x=0.5$ version of HYPOXIA.

- (1) Bayesian Defeatism is true (assumption for *reductio*).
- (2) If any form of Defeatism is true, your reduction in confidence in L from t_1 to t_2 is explained by a reduction of confidence in RatEL.
- (3) Such an explanation requires the truth of BRIDGE (or something like it).
- (4) If BRIDGE is true, then if (as Defeatism implies) your credence in L should be 0.5 at t_2 , your credence in RatEL should also be 0.5 at t_2 .
 - The argument for (4) is given in the Appendix, and depends on certain tricky auxiliary assumptions which we can discuss if there's time. Things would be a bit less straightforward if we had chosen $x > 0.5$.
- (5) So: your credence in RatEL should be 0.5 at t_2 (1–4)

- (6) **Key premise:** If you should have credence 0.5 in RatEL at t_2 , there must be a principled answer to the question why this is the case, and the only such principled answer is that 0.5 was your *prior* credence in RatEL (at t_0).
- A simple intuition behind this premise: the totality of what you've learnt since t_0 seems neutral as regards RatEL, just as it obviously would be if you had first seen someone *else* seeming to reason carefully from E to high confidence in L and subsequently got compelling evidence that they were hypoxic and actually only going through a simulacrum of "reasoning".
 - Note that for the argument below to go through, all we actually need is that credence in RatEL at t_0 wasn't *high*.
- (7) You should have credence 0.5 in RatEL at t_0 (5, 6).
- (8) If BRIDGE is true and your credence in RatEL should be 0.5 at t_0 , your credence in L given E should be 0.5 at t_0 .
- This also depends on the auxiliary assumptions from the Appendix.
- (9) Your credence in L given E should be 0.5 at t_0 (1–3, 7, 8).
- (10) RatEL is false (9, definition of RatEL)
- (11) If the premises of the above argument for (10) are true, you should have high credence at t_0 in the conjunction of them, and hence should have low credence in RatEL at t_0 .
- (12) $\perp$ (7, 11 — since 0.5 is not low).

4. An alternative model for the Bayesian defeatist

To avoid this, Miriam suggests that the Bayesian defeatist will need to say that RatEL is false: at t_0 , your credence in L conditional on E should not be high. Moreover, your credence in it should be low throughout.

- So, premise (1) is false. Premise (3) also turns out to be false given how Miriam develops this approach.
- This means that your high credence in L at t_1 can't be the result of conditionalizing on E. As we already suggested, the Bayesian Defeatist is going to have to attribute this high credence to conditionalizing on some further proposition J that you learn at the same time as E.
- *Toy model:* you yourself are an ignoramus to whom all these gauges and dials don't mean much. But you carry around a homunculus who is pretty good at reading gauges and dials, and which delivers a "judgment" to you after you have looked at

them. You are initially confident that the homunculus won't make a mistake, so conditionalizing on "the homunculus judged L" leaves you confident in L.

- Notice that on this model, even if on some occasion your homunculus malfunctioned and issued a judgment completely at random, leading you (by conditionalization) to have high confidence in the proposition judged, that won't detract from *your* rationality at the time in question. (Of course, the more this happens, the less you should trust your homunculus.)

Miriam makes two objections to this alternative model.

1. On this model, even if learning H reduces your credence in L to 0.5, it does not support the claim that your t_1 -credences concerning L were "rationally subpar". On the plausible model of how hypoxia tends to lead people to have high credence in falsehoods, its main effect is on the mechanism (your homunculus) that generates the "judgment", J, not on your ability to conditionalize on J. So 'the "higher order" aspect of higher order defeat seems to fall away' (13) — the defeat does not involve thinking about the possibility of your own (past) irrationality.
 - I think the proponent of the view might respond to this by saying that in a broad sense of 'rationality', one can count as 'irrational' in virtue of having a malfunctioning homunculus, even if one responds perfectly to the homunculus's outputs. In this broad sense, the bad effects of hypoxia do involve making people irrational.
2. 'It also makes the very process of reasoning mysterious. We appear to be trying to figure out what E supports, but if we know that E supports nothing, what should we take ourselves to be doing?' (13)
 - I agree with this objection. One might say that the whole point of the Bayesian machinery is to give an account of how one's credences should be sensitive to one's evidence — it is pointless if we have to outsource the central job to a homunculus (which isn't itself modeled in Bayesian terms).

5. Can we give a more broadly Bayesian version of defeatism?

We have been assuming a fairly demanding interpretation of Bayesian defeatism, on which it requires one to explain the relevant credential transitions—the increase in your credence in L from t_0 to t_1 and its decrease from t_1 to t_2 —by appeal to *conditionalization*. But might there be a less demanding way to develop a defeatist view that's still in some way recognizably Bayesian?

- A familiar point in the literature on 'bounded Bayesian rationality' is that conditionalization is computationally intractable. More precisely: it's computationally intractable when you're conditionalizing on is *rich, detailed* evidence (like what you get

from your dials, gauges, and maps), and the internal representation of the prior conditional credences is highly *implicit*.

- One might—very much in the spirit of contemporary Bayesian epistemology—appeal to considerations of *accuracy* in explaining what makes the difference between good and bad approximation methods. Of course, a method which guaranteed that you would conditionalize would have maximal accuracy; but this method will get excluded as ‘unfeasible’.
- Among the ‘feasible’ methods, we could consider methods which, when they go well, lead to credence functions such that (i) conditional on ‘I functioned well’, closely approximate the result of conditionalizing on your priors + your new evidence + ‘I functioned well’, but (ii) conditional on ‘I malfunctioned’, are closer to the result of just conditionalizing your priors just on ‘I malfunctioned’ (at least as regards propositions far downstream of your evidence).
- Plausibly, methods that lead to credence functions with properties (i) and (ii) are going to have distinctive accuracy-theoretic advantages. A key advantage of such methods is that *even when you malfunction*, your resulting credence function will still typically have property (ii), which means you’ll be more accurate if you later learn that you malfunctioned.
- If so, there’ll be an accuracy-theoretic explanation of why it’s best (given your human limitations) to have the sort of approximation method that, typically, will lead you after learning E to have a credence function that approximates $\Pr(\neg H)\Pr(\cdot | E \neg H) + \Pr(H)\Pr(\cdot | H)$, at least as regards propositions far downstream of E.

There’s a further step from this to the letter of defeatism (‘your credence in L at t_2 should be x ’). It’s not obvious how we ever get to such ‘should’ claims out of evaluative claims about general dispositions. We might say ‘the credences you *should* have are the ones you would have if you had the best dispositions and they functioned as well as possible’. But it’s unclear what it is for them to ‘function as well as possible’—after all, the goodness of the dispositions turns on facts about the shape of the whole spectrum of ways they can function.

Nevertheless, I am optimistic that the ‘bounded rationality’ perspective can provide a satisfying account of the defeatist intuitions.

The auxiliary assumptions (from Miriam’s appendix)

Assumption 0 (stipulated in the case?): for some $r \gg 0.5$, you are certain throughout that where p is the RUP, either $p(L | E) = r$ (and thus RatEL) or $p(\neg L | E) = r$ (and thus $\neg$ RatEL).

Assumption 1: for some $r^* \gg 0.5$, you are certain throughout that where p is the RUP, either $p(L | E) = r$ and $p(L | E \wedge p \text{ is the RUP}) = r^*$, or $p(\neg L | E) = r$ and $p(\neg L | E \wedge p \text{ is the RUP}) = r^*$.

Assumption 2: you are certain throughout that where p is the RUP, $p(L | E \wedge H \wedge p \text{ is the RUP}) = p(L | E \wedge p \text{ is the RUP})$. (I.e., the relevance of H is screened off by $p \text{ is the RUP}$).

Assumption 3 (???): you are certain throughout that any extra evidence J you got at t_1 is irrelevant to L given E and the truth about rationality: where p is the RUP, $p(L | E \wedge p \text{ is the RUP}) = p(L | E \wedge p \text{ is the RUP} \wedge J)$

Then we have $0.5 = P_2(L) = P_0(L | E \wedge J \wedge H) = \sum_q P_0(q \text{ is the RUP} | E \wedge J \wedge H) q(L | q \text{ is the RUP} \wedge E \wedge J \wedge H) = \sum_q P_0(q \text{ is the RUP} | E \wedge J \wedge H) q(L | q \text{ is the RUP} \wedge E \wedge J) = \sum_q P_0(q \text{ is the RUP} | E \wedge J \wedge H) q(L | q \text{ is the RUP} \wedge E) = r^* P_0(\text{RatEL} | E \wedge J \wedge H) + (1-r^*) P_0(\neg \text{RatEL} | E \wedge J \wedge H)$. So $P_0(\text{RatEL} | E \wedge J \wedge H) = P_2(\text{RatEL}) = 0.5$.